

The Small World of the Amarna Letters, Revisited: A Reproducible, Bias-Aware Network Analysis

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All numbers regenerate from `run_all.sh`; seeds fixed. Target venue: *Journal of Historical Network Research*.

Abstract

The Amarna letters (c. 1360–1330 BC) are the oldest well-preserved archive of interstate diplomacy, and the network analysis of [Cline and Cline \(2015\)](#) — a hand-coded “small world” with clustering $\sim 49\times$ a random graph, brokered above all by Rib-Hadda of Byblos — has become the canonical quantitative reading of the corpus. We rebuild the Amarna networks from the Oracc lemmatized edition with fully reproducible code, hand-coding only what the digital corpus lacks, and subject every headline claim to appropriate null models and survival-bias corrections. Three results follow. (1) The correspondence network contains *zero* triangles: the celebrated small world lives entirely in the mention layer. Reconstructed at full scale (244 persons; Cline & Cline counted 246), that layer’s clustering ($0.409 \approx$ their 0.391) exceeds a fair bipartite configuration null by a factor of ~ 1.6 — genuine structure, but far from the “ $48.75\times$ ” figure produced by an Erdős–Rényi comparison. (2) Rib-Hadda’s famous brokerage replicates exactly under Cline & Cline’s construction and dissolves under bias correction: it depends on splitting the pharaoh’s identity across nodes and on his 60+-letter dossier; volume-equalized, he falls to median rank 15 with zero bootstrap probability of a top-three position, while Aziru of Amurru and the Egyptian commissioner Yanhamu emerge as the stable brokers. (3) Inferential models formalize the tier structure (ERGM: reciprocity $+3.78$, tier anti-homophily -2.21 , both $p < 0.001$), and geography splits cleanly by layer: distance decays mention and conflict ties — neighbors name and accuse neighbors — while epistolary volume to the court shows no robust distance effect, a contrast that survives compositional drops and control for proximity to Egyptian administrative centers. We release the actor registry, edge lists, provenance joins (including a machine-readable transcription of the petrographic determinations of [Goren et al., 2004](#)), and all code.

1 Introduction

The 382 tablets found at Tell el-Amarna record the diplomatic correspondence of the Egyptian court under Amenhotep III and Akhenaten with two tiers of partners: the “Great Powers” (Babylonia, Assyria, Mittani, Hatti, Arzawa, Alashiya) and the vassal rulers of the Levant ([Moran, 1992](#); [Rainey, 2015](#); [Cohen and Westbrook, 2000](#)). [Cline and Cline \(2015\)](#) introduced network analysis to this corpus, hand-coding actors and ties in NodeXL, and reported three memorable findings: a clustering coefficient of 0.391 , “nearly fifty times higher” than a comparable random network; the minor king Rib-Hadda of Byblos out-scoring both pharaohs in betweenness centrality; and pharaohs dominating only two of ten network clusters. [Cline \(2025\)](#) brought this picture — 246 named people, 464 connections — to a broad audience.

These studies are exploratory: no null models, no uncertainty, no correction for the archive’s structure, and hand-coded data that cannot be audited. The present study asks which of their find-

ings survive reproducible construction and appropriate inference — and finds that the answer is a precise decomposition rather than a verdict: each headline claim is true under one specific set of constructional choices and false under defensible alternatives. Identifying *which* choice drives *which* result is, we argue, the useful contribution.

A second contribution is methodological framing, in the tradition of archaeological network science’s missing-data literature (Collar et al., 2015; Brughmans and Peeples, 2023). The archive is an ego-network of Egypt, excavated in Egypt: every correspondence tie touches the court by construction, and every mention survives because some letter to Egypt carried it. Centrality measured on such an archive is *salience to Egypt*, not historical importance. Rather than treating this as a caveat, we build it into the analysis: the correspondence layer is analyzed as what it is (a star), the substantive questions are pushed to the mention and conflict layers, and every centrality claim carries bootstrap rank intervals and dossier-volume corrections.

2 Data

Corpus. The Oracc aemw/amarna edition (Lauinger and Yoder, 2025, catalogue timestamp 2024-07-05) contains 379 catalogued tablets, of which 347 are letters and 305 carry lemmatized transliteration (34,346 word tokens, 94.0% lemmatized). The Great Powers letters EA 1–44 and ~30 fragments have catalogue metadata but no digital text.

A correction to the record. Contrary to the assumption (reproduced in our own project plan from the Oracc pager configuration) that the catalogue lacks structured sender/addressee fields, the raw catalogue.json carries `ancient_author` for 349 and `recipient` for 348 of 379 entries — including 42/44 of the unlemmatized Great Powers letters. We therefore used the catalogue fields as the primary source and demoted formula parsing to a validation instrument.

Validation. A parser extracting sender and addressee from the stereotyped address formulae (cf. Mynářová, 2007) agrees with the catalogue on 96.1% (sender) and 96.5% (addressee) of comparable letters. All 18 disagreements were adjudicated against both English editions (Moran, 1992; Rainey, 2015). Both editions confirm two catalogue errors (EA 62 is addressed to the commissioner Paḥanate, not pharaoh; EA 301 is from Šubandu, not Yapahu), as well as EA 7’s recipient (“Babylon” corrected to the pharaoh Rainey restores as [Naphu]rureia). Twelve cases resolve for the catalogue with both editions agreeing; three remain edition-contested and are graded *disputed*. Every verdict, with supporting quotations from both editions, ships in the repository.

Entity resolution. Actors are keyed on alias-canonicalized folded names; the hand-curated registry (17 entries) records Sumerographic readings (IR₃-Ḫebat = ‘Abdi-Ḫeba), orthographic systems (Oracc ‘Adiri = conventional Aziru), contested sign readings, and pharaonic throne names, each with a justification. All named pharaohs collapse to a single PHARAOH node in the base analysis; the per-letter identification is retained for sensitivity splits.

Hand-coded supplements. EA 1–44’s mentioned persons were coded from Moran’s translations (44 letters; gods excluded, throne names resolved to PHARAOH). A signed conflict layer (accusation/alliance) was coded via a corpus-wide hostile-language sweep of Moran’s translations (40 letters, 63 edges: 59 accusations, 4 alliances), each edge carrying its Moran quotation. The full set was then independently recoded from Rainey’s translations: cross-edition agreement is 93.5% with Cohen’s

$\kappa = 0.86$, all divergences traceable to textual differences between the editions rather than to the coding rules. This establishes edition-robustness; since both passes share an annotator, an independent human recoding remains the outstanding reliability step.

Geography and provenance. Politics were georeferenced against Pleiades (31/55 automated; regions without ancient fixed points remain unlocated by design), and every coordinate entering a distance test was hand-verified against the Pleiades bulk dump — a necessary step, since classical gazetteers are homonym-rich (Pella, Arca, Tyros, and “Syria” all have distant namesakes). The petrographic provenance determinations of Goren et al. (2004) were transcribed to a machine-readable table (292 tablets); where both sources speak, the petrographic determination agrees with the catalogue’s provenience 87% of the time, and the disagreements reproduce Goren et al.’s own discoveries.

3 Network construction

Three layers, all built by code from the tables above:

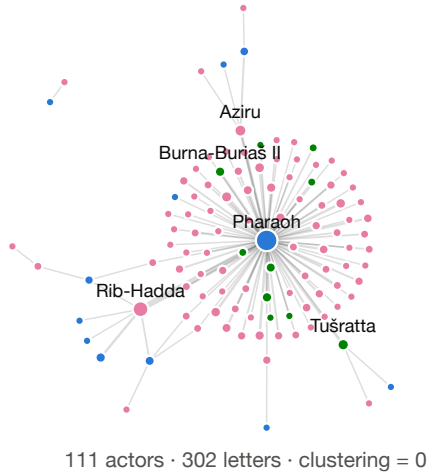
1. **Correspondence** (directed, weighted): sender $\rightarrow$ addressee per letter; 111 actors, 302 letters, 117 distinct dyads, every endpoint carrying a confidence grade.
2. **Mention co-occurrence** (undirected, weighted): the persons appearing in each letter form a clique; 244 persons, 734 dyads. The Great-Power letters enter via the Moran hand-coding.
3. **Conflict** (signed, directed; provisional): 63 accusation (−) and alliance (+) edges across 40 letters, with quotation-level provenance.

We additionally rebuild a **Cline-style graph** — correspondence edges plus sender $\rightarrow$ mentioned stars — because replication requires the original’s construction, not just its corpus.

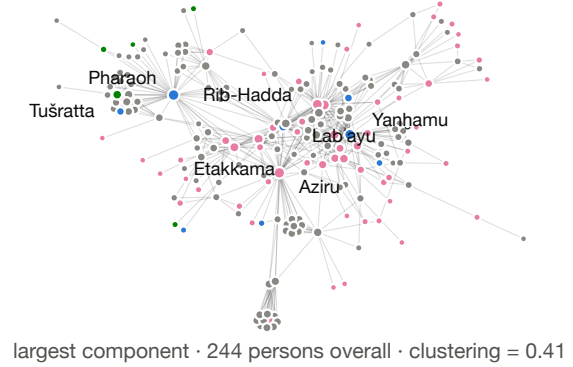
4 Methods

Descriptives are never compared to an Erdős–Rényi baseline alone. The correspondence and Cline-style graphs are tested against degree-preserving rewiring; the clique-projected mention layer against a *bipartite* configuration null (shuffling person-to-letter incidences while preserving letter sizes and person frequencies), since clique projection manufactures triangles that edge-rewiring nulls do not discount (Brughmans and Peebles, 2023). Centrality claims carry (a) bootstrap-over-letters 95% rank intervals (competition-ranked under ties) and (b) a dossier-equalization design in which every sender contributes at most the median dossier size (2 letters) per replicate. Inference: an ERGM on the binarized correspondence network (Hunter and Handcock, 2006; Handcock et al., 2008), with a Bayesian replication (Caimo and Friel, 2011); a latent-space cluster model on the mention layer (Krivitsky and Handcock, 2008); community detection by both Louvain and Leiden (Blondel et al., 2008; Traag et al., 2019); and QAP/permutation tests (Freedman and Lane, 1983) for the geography contrasts. All analyses are seeded; `run_all.sh` regenerates every number in this paper.

(a) Correspondence: who wrote to whom



(b) Mention: who is named together



● Egypt (court & officials) ● Great Powers ● Vassals ● unknown

Figure 1: The two layers of the Amarna network. (a) The correspondence layer is a star on the court: everyone writes to Egypt, and its global clustering is exactly zero. (b) The mention layer — persons named together in the same letter — is where all triadic structure lives. Node area scales with letter volume (a) and degree (b); layouts are seeded Fruchterman–Reingold (Csárdi and Nepusz, 2006).

5 Results

5.1 The correspondence network has no small world to find

Global and local clustering in the correspondence layer are exactly zero in all three edge sets (all letters; high-confidence only; without the Byblos dossier) — below even the configuration null (95% band 0.0015–0.0051). The layer is a star on Egypt (Figure 1a): vassals and kings wrote to the court, not to each other (two vassal–vassal dyads exist in 302 letters). Whatever the “small world of the Amarna letters” is, it is not a property of who wrote to whom.

5.2 The small world is real, mention-borne, and $\sim 1.6\times$ — not $49\times$

Rebuilt at full scale, the person-to-person mention network has 244 persons (Cline & Cline: 246 people) and clustering 0.409 (theirs: 0.391) — we take this convergence as evidence that we have reconstructed their object. An Erdős–Rényi comparison of the kind behind the “ $48.75\times$ ” claim yields $16.5\times$ here. But against the bipartite configuration null the observed 0.409 stands over a null median of 0.253 (95% band 0.235–0.271): significant — the observed value lies outside the null band in every variant, including without the Byblos dossier (0.452 vs. 0.315) — but a factor of ~ 1.6 (Figure 2). The small world of the Amarna letters exists; the astronomical multiplier was an artifact of the weakest available null. The Cline-style star construction, by contrast, clusters at 0.034, at or below its null — confirming that triadic structure comes from mentions connecting *third parties*, not from anyone’s correspondence.

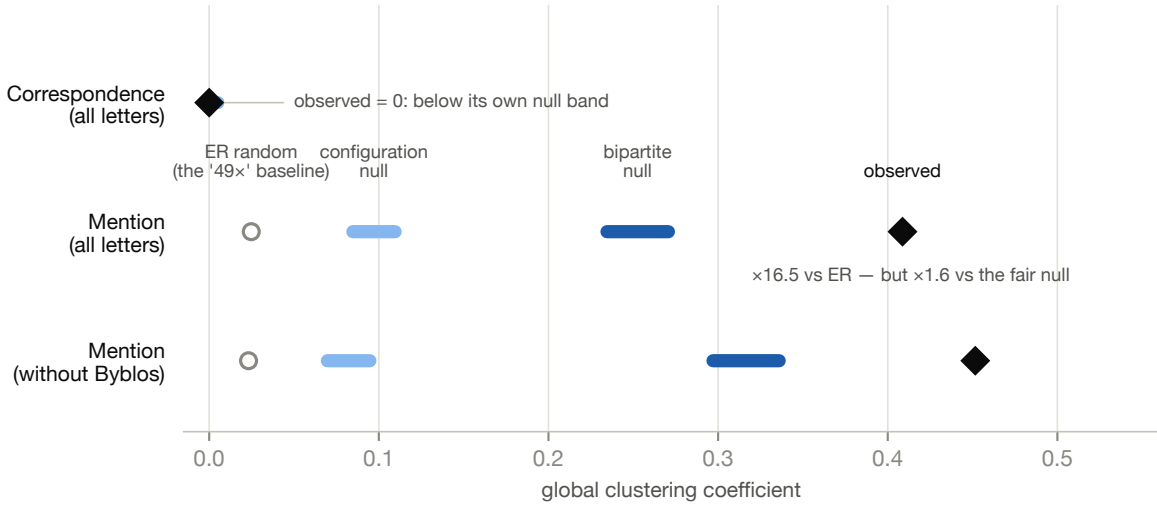


Figure 2: Observed clustering against three null models. The multiplier one reports is a choice of null: against Erdős–Rényi the mention layer is $\sim 16\times$ “random”; against a bipartite configuration null that knows letters come in groups, $\sim 1.6\times$. The correspondence layer sits *below* its own null band at exactly zero.

5.3 Rib-Hadda’s brokerage, decomposed

Cline & Cline’s flagship finding — Rib-Hadda of Byblos out-brokering the pharaohs — replicates *exactly* under their choices: in the Cline-style graph with pharaohs split by identification, Rib-Hadda’s betweenness exceeds Amenhotep III’s and Amenhotep IV’s. But the residual unidentified pharaoh — most letters address only “the king” — holds roughly five times Rib-Hadda’s score. The finding is an artifact of dividing one actor’s identity across nodes under uncertainty. From there the correction proceeds stepwise (Figure 3): collapse the pharaohs and Rib-Hadda is second; move to the person-to-person layer and he is ~ 7 th (bootstrap median 7, 95% interval 4–12); equalize dossier volume and he is 15th with $P(\text{top } 3) = 0.00$. His 63-letter dossier buys eigenvector attachment to the hub, and nothing else.

The stable brokers, in every construction and correction, are **Aziru of Amurru** (rank 1 with $P(\text{top } 3) = 0.94\text{--}0.98$ across designs) — the defector who bridged Egypt’s orbit and Hatti’s, and simultaneously the most-accused actor in the conflict layer (Figure 4) — and, once volume is corrected, **Yanhamu**, the Egyptian commissioner ($P(\text{top } 3) = 0.66$), vindicating the hypothesis that pharaoh’s officials, not the loudest vassal, carried the network. Rib-Hadda’s fame is a fact about his archive, not his position.

5.4 The diplomatic system as coefficients

The ERGM formalizes the two-tier reading (cf. [Liverani, 2001](#); [Podany, 2010](#)): reciprocity is strong and significant (mutual = $+3.78 \pm 0.80$), concentrated at the Great-Power tier (17% of Egypt↔Great-Power dyads reciprocated vs. 4% of Egypt↔vassal, the latter being Egyptian file copies), and tier homophily is strongly *negative* (nodematch = -2.21 ± 0.64): Amarna correspondence is cross-tier by construction — the brotherhood of kings and the servitude of vassals both materialize as letters to Egypt, not within-tier correspondence (Table 1). The Bayesian replication agrees (mutual 95% credible

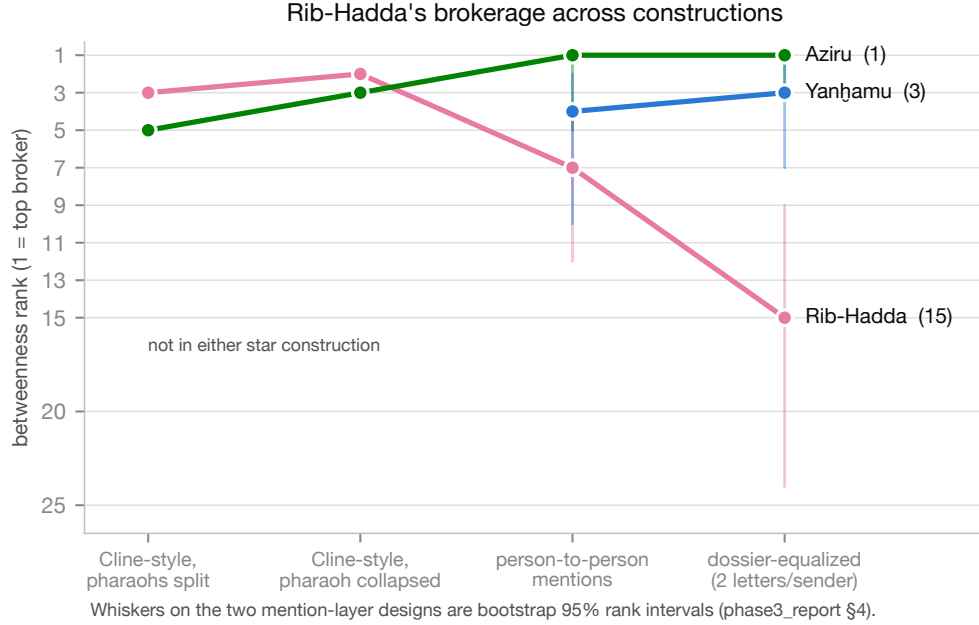


Figure 3: Betweenness rank of three actors across four constructions, from Cline & Cline’s split-pharaoh graph (left) to the dossier-equalized mention layer (right). Rib-Hadda’s celebrated brokerage exists only in the star constructions; Aziru holds rank 1 in every corrected design, and the commissioner Yanhamu — invisible in either star construction — rises once volume is equalized.

interval $[2.3, 6.2]$; nodematch $[-4.0, -0.8]$). The latent-space model finds clusters that do not recover the tiers: mention-layer communities (modularity ≈ 0.56) are political theatres — the Amurru crisis, the Damascus–Qadesh axis, the Shechem–Jerusalem conflict, the Tyre–Sidon feud — not diplomatic ranks.

5.5 Geography: distance structures mentions and conflict, not correspondence

Distance decays ties in the mention layer (point-biserial $r = -0.147$, QAP $p < 0.001$): neighbors name neighbors. The conflict layer agrees ($r = -0.298$, QAP $p = 0.001$, $n = 14$ located actors), and drawing the conflict network in space makes the result visible (Figure 6): accusations run be-

Table 1: ERGM on the binarized correspondence network (MCMLE; GOF adequate on all model statistics, MC $p \geq 0.94$). The Bayesian replication (Bergm, 30k iterations) gives concordant posteriors.

term	est.	s.e.	p
edges	-2.58	0.68	0.0002
mutual (reciprocity)	+3.78	0.80	< 0.0001
receiver: Great Power	-4.20	0.77	< 0.0001
receiver: vassal	-4.07	0.53	< 0.0001
sender: Great Power	+0.12	0.73	0.86
sender: vassal	+0.11	0.68	0.87
tier homophily (nodematch)	-2.21	0.64	0.0006

tween neighbors — along the Phoenician coast, across the Beqaa, and within the crowded hill country between Gezer, Gath, Shechem, and Jerusalem — while the one long-range structure is the Qadesh–Damascus axis of the Hittite crisis. Correspondence with the court behaves differently. The raw correlation between a vassal’s letter count and distance from Akhetaten is $\rho = -0.32$ ($p = 0.07$), and it is not robust: a leave-one-polity-out jackknife retains nominal significance in only 7 of 24 deletions; dropping the polities with political reasons to write little — Hatti-aligned Qadesh and Sidon (EA 174–176, 148–149) and quasi-independent Ugarit — leaves $\rho = -0.18$ ($p = 0.32$); and in a partialled design against hand-coded covariates — distance to the nearest Egyptian administrative center (Gaza, Joppa, Beth-Shean, Kumidi, Şumur; Figure 5; cf. Morris, 2005), coastal access, Hatti alignment — nothing predicts epistolary volume, court distance included (partial $r = -0.26$, $p = 0.17$). The sign stays negative in every variant, so a weak true decay is not excluded; but the defensible statement is the pre-registered one: geography structures whom the letters *talk about*, not who talks to Egypt. (The repository documents how an apparently strong correspondence decay in an earlier draft traced to gazetteer homonym mismatches and a rank-tie artifact — the coordinate verification described in §2 exists because of that episode.)

The signed conflict network (hand-coded, 40 letters)

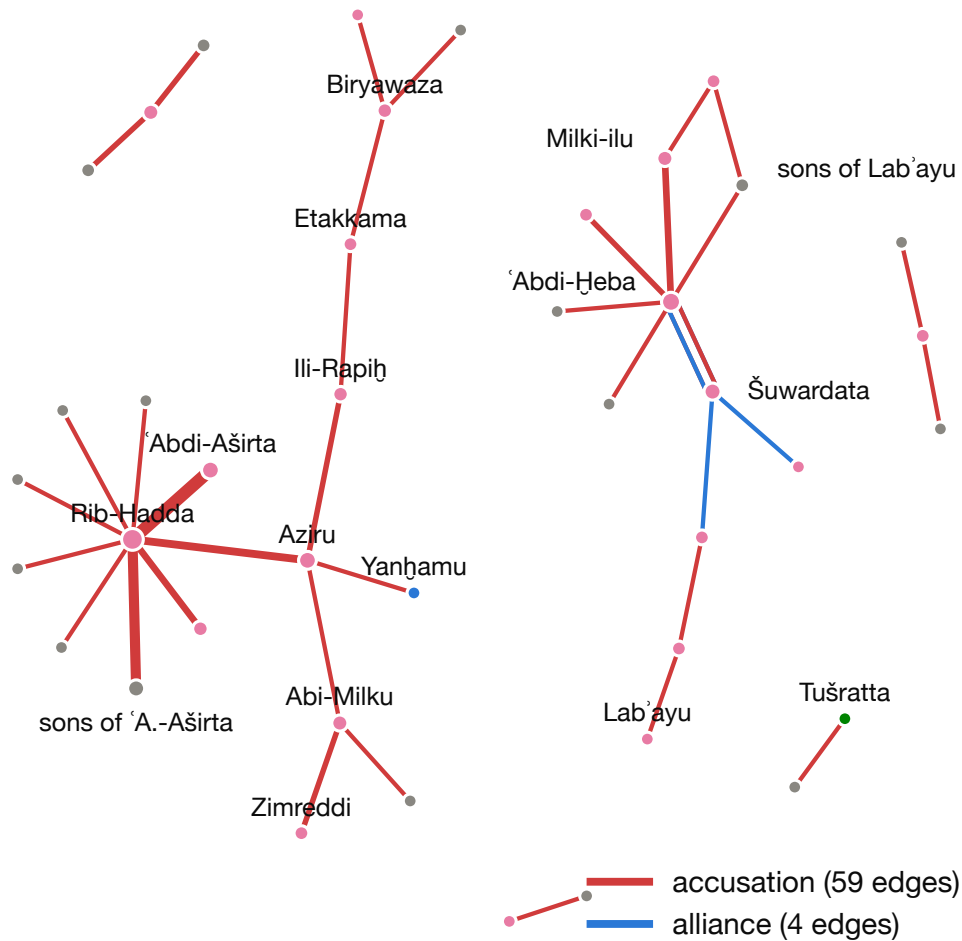


Figure 4: The signed conflict network, hand-coded from hostile and allied language (edges drawn undirected; both an accusation and an alliance connect Šuwardata and 'Abdi-Ḫeba, EA 366). The regional theatres — Amurru (left), Damascus-Qadesh (center), Shechem-Jerusalem (right) — are invisible in the correspondence layer.

The corrected geography of the corpus

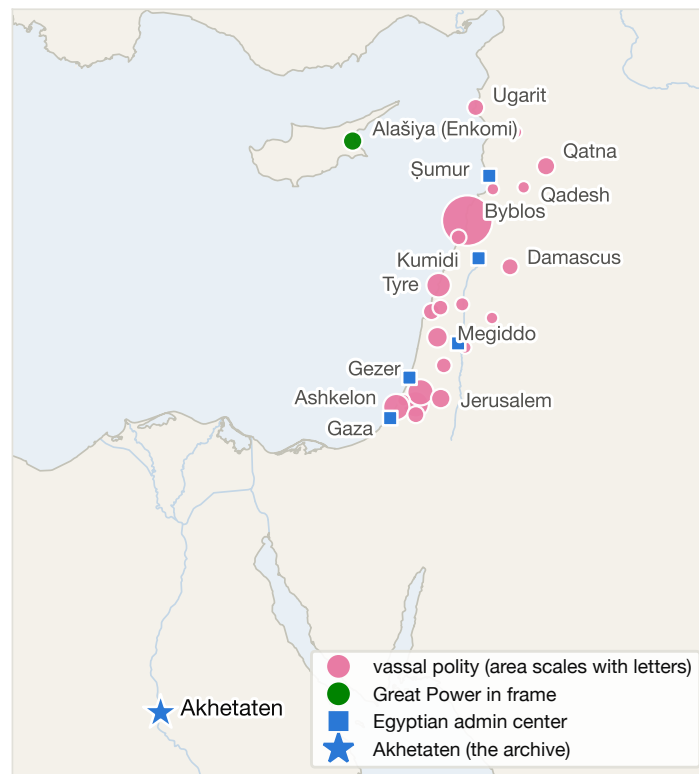
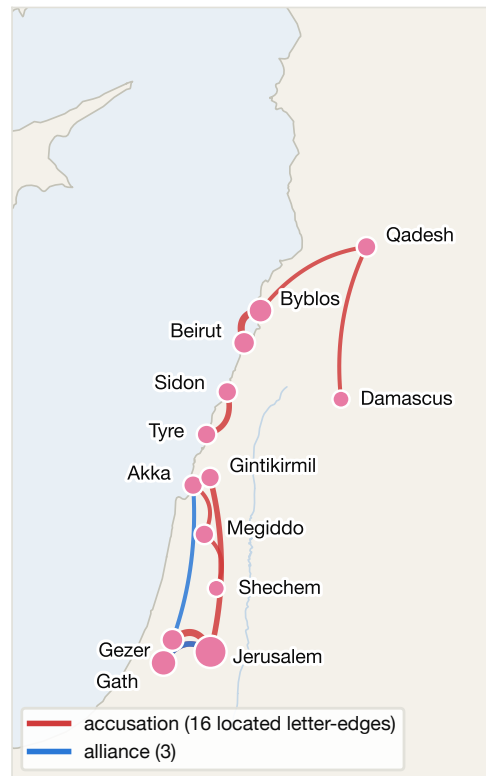


Figure 5: The corrected geography of the corpus: located vassal polities (area scales with letters to pharaoh), the five Egyptian garrison/commissioner seats used as the administrative-density covariate, and Akhetaten. Region-states without fixed points (Amurru, Mittani, “Syria”) are excluded rather than imputed.

The conflict network in space



Curved lines connect the polities of accuser and accused; width scales with letters. Edges involving unlocated actors (collectives, the 'Apiru, fragmentary names) are not drawn.

Figure 6: The conflict network in space: accusation (red) and alliance (blue) edges between located polities, aggregated over letters. Accusations run between neighbors — the Phoenician coast, the Gezer–Gath–Shechem–Jerusalem hill country — with the Qadesh–Damascus axis of the Hittite crisis as the one long-range structure. Edges involving unlocated actors (collectives, the 'Apiru) are omitted.

6 Discussion

The revision this study proposes is not that Cline & Cline were wrong but that each of their results is the output of an unstated constructional switch. Split the pharaoh’s identity and Rib-Hadda out-brokers kings; collapse it and he does not. Compare against an Erdős–Rényi graph and the archive is a $49\times$ marvel; compare against a null that knows letters come in groups and it is a $1.6\times$ society. Count a man’s own dossier and he is central; ask what remains when every voice speaks at equal volume and the brokers are the defector Amurru dynast and pharaoh’s own commissioner. None of these switches is illegitimate — but they must be visible, and their consequences measured, before centrality in a survival-biased ego-archive is read as ancient history.

What survives every correction is historically satisfying: a two-tier system statistically legible in one coefficient (+ reciprocity above, – homophily across), inter-vassal politics organized in regional conflict theatres invisible to the correspondence graph, brokerage lodged in exactly the two figures — Aziru, Yanḫamu — whose careers the letters narrate as boundary-crossing, and a mention society whose modest but real small-world excess reflects genuine third-party political entanglement. The archive measures salience to Egypt; measured carefully, that salience still has structure worth explaining.

7 Limitations

The conflict layer is corpus-wide and edition-robust (cross-edition $\kappa = 0.86$) but single-annotator; an independent human recoding remains desirable. EA 1–44 mentions were hand-coded from translation rather than lemmata (a documented source seam that will close when Oracc releases the Great Powers letters). Senders whose polity the catalogue string does not state default to the vassal tier (14 minor correspondents of 1–3 letters each); Ugarit’s rulers are likewise coded vassal though the kingdom’s standing was closer to client-of-both-courts. At these actors’ letter volumes neither choice can move the tier coefficients, but both are choices, not facts. Chronology is coarse by design: a two-phase sensitivity split reproduces the small-world excess within each era and keeps brokerage with the Amurru dynasty in both — ‘Abdi-Aširta early, Aziru late. Geographic coverage reaches 64% of letters at both endpoints; the excluded region-states include Amurru, so the compositional analysis of §5.5 cannot see the system’s most consequential defector.

8 Data and code availability

All code (MIT), derived tables (CC-BY), figures, and this draft: <https://git.sr.ht/~calgacus/ussira-pitati>. Oracc corpus text is CC-BY-SA (Lauinger and Yoder, 2025) and is not redistributed; Moran (1992) and Goren et al. (2004) are quoted within scholarly fair use. A Zenodo deposit with DOI accompanies submission.

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